How many Different types of data/ operand

↳ 4 → number
→ char
→ logic

Addressing → type (reg + Memory)
→ mode
→ Actual Address
↳ effective "
(logical/ virtual Format)
↓
unsigned integers

number → signed
unsigned
float

packed
↓
Binary rep.

range limitation

precision limitation

Conversion

Arithmetic → Unary
→ Binary

flags → reflect particular operation

ALU required.
(shift)
rotate

logical

Data Transfer → reg mode

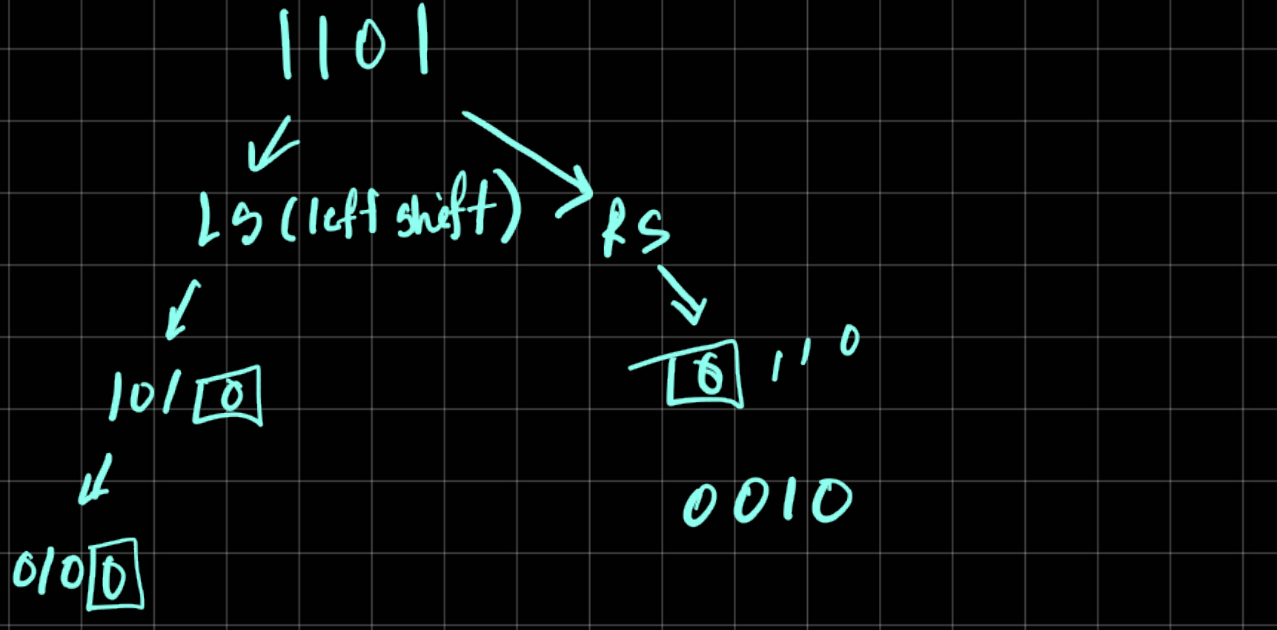
→ Memory → EA → PA → Cache → Memory.

I/O conversion

system control

Transfer "

rotate "10
right rotate

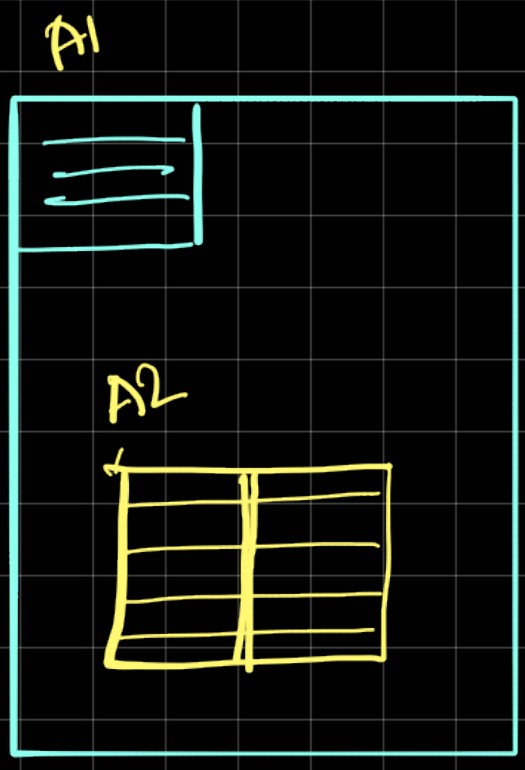


Shift → sign bit unchanged

BCD → 42 - 2-1

convert → easy method

use mapping; keep table ...
change the corresponding val.



To A1(L), A2

Target address.

Transfer

Branch (x)

Jump x

Break

Prog

backward

skip

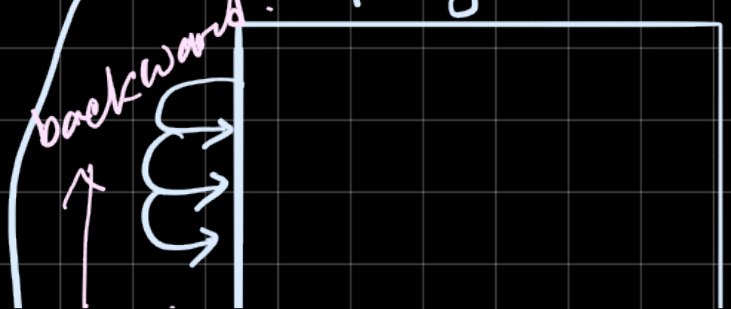
Call

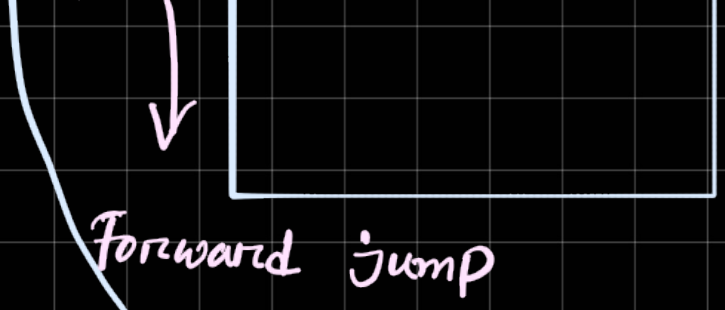
call

return

program

save (change PC)





→ if else condⁿ

Loop.

Branch 202 → unconditional.

0 1-1 → jump to address 211

X = X - Y

:

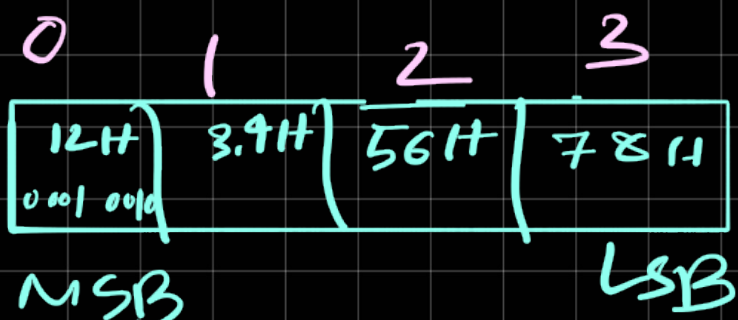
(-1)

Proc 2

pc holding 4601

↳ return back to Proc 1.

Storing address prblm → use stack.



Word Array

184

For byte Address → 4 byte
other option

1 option

12 = MSB 184

185

186

187 78 = LSB

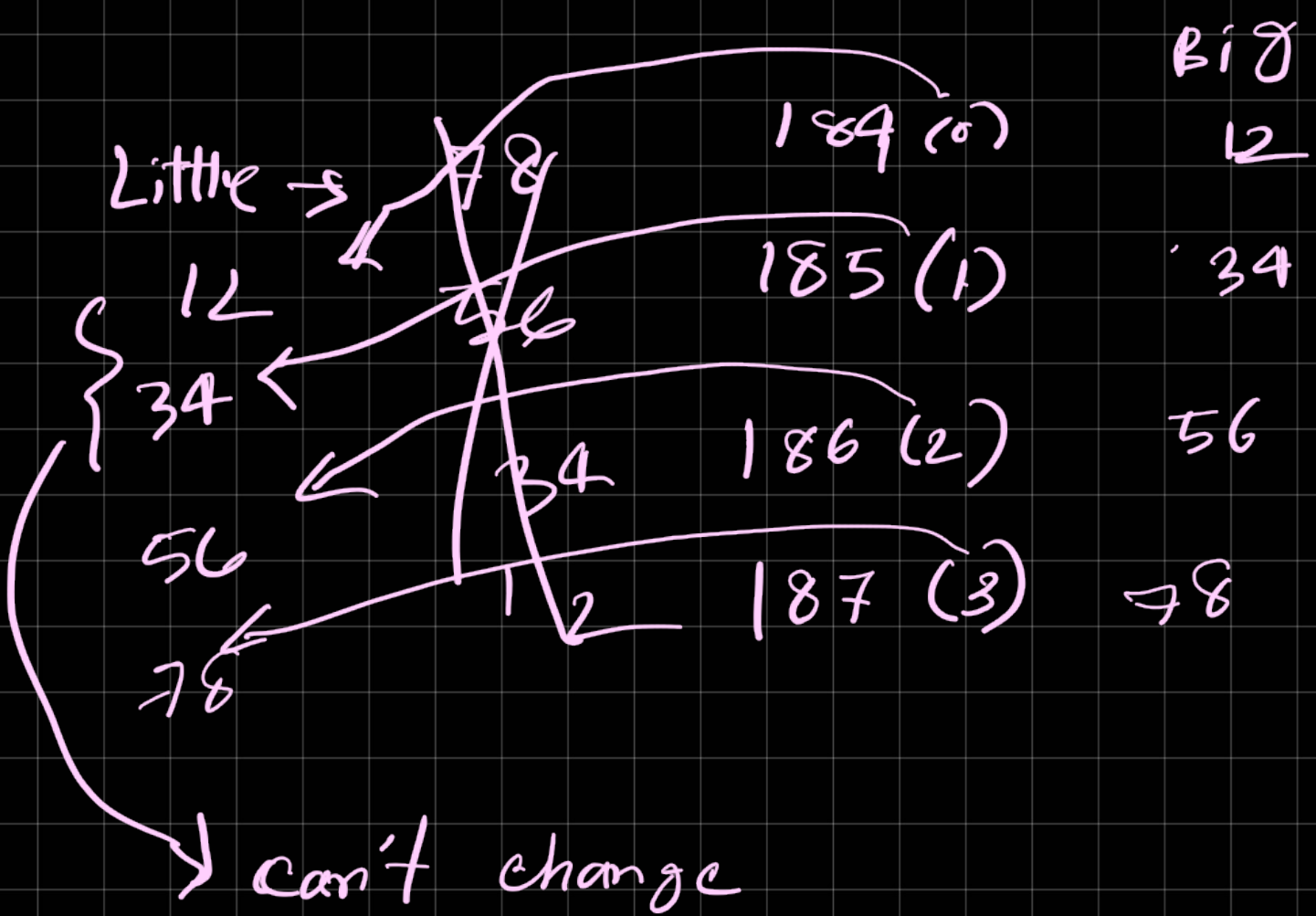
MSB = 12

Data is fragmented.

Big

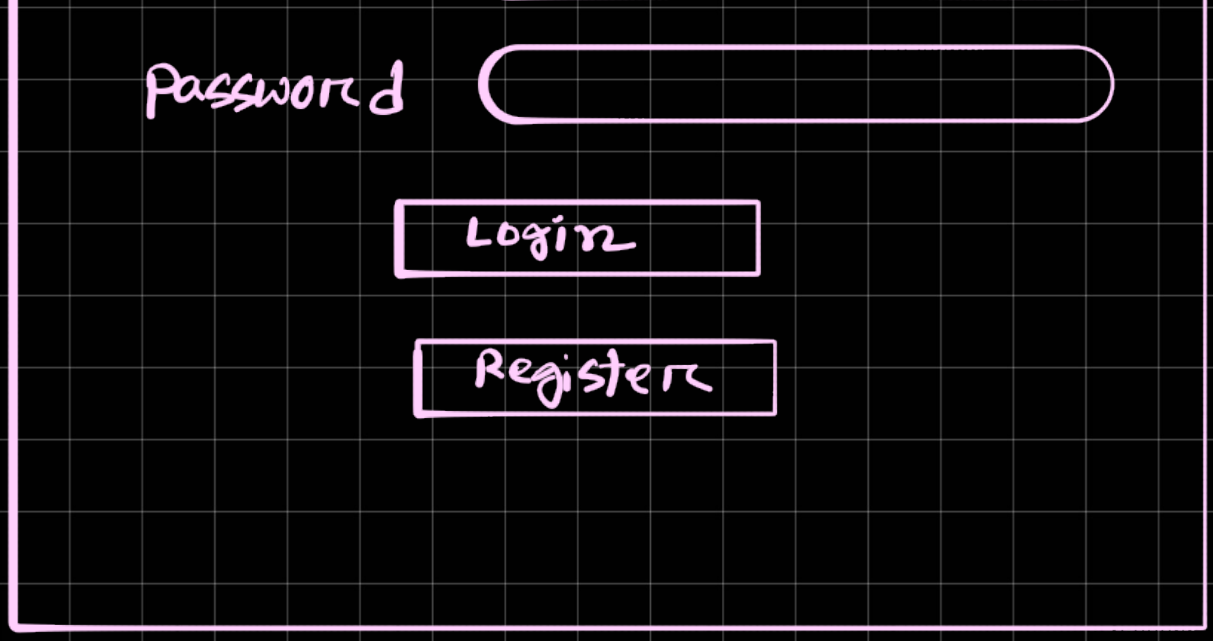
Little

For 0 1 2 3 → Little



Ch 12 end.

User Name


Password
Login
Register